Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/13/2026
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Single-Source Electromagnetic Levitation Flight Support for Ternary Alloy Rapid Solidification (ELFSTARS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	04/26/2024	**End Date:**	04/25/2026
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nawer, Jannatun Ph.D. (Boise State University)		
Grant/Contract No.:	80NSSC24K1119		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This research addresses fundamental issues relating to modeling alloy behavior prediction, phase selection, and analysis of the processes governing microstructural evolution, which lead to improved modeling of Additive Manufacturing (AM) processes in space, low gravity, and on Earth.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Graduated 1 PhD student, Colby Azersky, now working at NASA Goddard Space Flight Center. Initiated research activities supporting 2 Tufts BS senior honors thesis students. Published 1 peer-reviewed article, 1 thesis submission, and 5 scientific talks. Sponsored student research projects using the NASA Electrostatic Levitator which culminated in in-person experimentation visits to MSFC by Tufts undergraduate students Brian Fu, Ben Fisher and graduate student Peace Muusha. Conducted International Space Station (ISS) expedition 71 co-investigation (TiAl#66Ar), remote operation testing, April 16-17, 2024. Attended the 40th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), San Juan, Puerto Rico, December 3-7, 2024, as organizer and as lead for the ISS-NL User Advisory Committee Public Meeting. Attended the 22nd National Institute of Standards and Technology (NIST) Properties for Materials Science at High Temperatures Symposium, Boulder CO, June 23-28, 2025 as organizer. Attended the Material Science Laboratory Electromagnetic Levitator (MSL-EML) Investigators Working Group (IWG) #23, at the European Space Agency (ESA) European Space Research and Technology Centre (ESTEC), Noordwijk Netherlands, 21-22 May 2024 as IWG chairman and organizer. Refined test plan documentation to run future Batch 3/4 alloys using ISS-EML. (FeCo30, FeNi10, HEA, FeCB, TiZrNi) with the German Space Agency (DLR) Microgravity User Support Center (MUSC). Obtained approval for test plan and delivered ground support and flight samples for Batch 4 (Inconel 626, Inconel 718, and Gold). Supported proposal collaborations for Batch 5 candidates as part of the ESA peer review process.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Abstracts for Journals and Proceedings	Matson D, Muusha P. "Advanced in-space inductive measurements of thermophysical properties using parabolic flight technology." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Matson D, Muusha P. "Inductive measurement of electrical resistivity and density of Fe90Ni10, Fe72Cr13Ni15 and Fe72Cr17Ni11 using the sample coupling electronics on board the ISS International Space Station." 22nd Symposium on Thermophysical Properties, Boulder CO, June 26, 2024. Abstracts. 22nd Symposium on Thermophysical Properties, Boulder CO, June 26, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Matson D, Volkmann T. "ISS-EML batch #5 – MAGNEPHAS samples Fe80Ni20 (#143) and Fe99B1 (#144) & ELFSTONE STARS samples TA625-Mo (#145), TA625-Fe (#146), TA718 (#147) and TA718-Fe (#161)." Investigators Working Group Meeting – IWG#23, ESA-ESTEC, Noordwijk Netherlands, May 22, 2024. Abstracts. Investigators Working Group Meeting – IWG#23, ESA-ESTEC, Noordwijk Netherlands, May 22, 2024. , May-2024
Abstracts for Journals and Proceedings	Matson D, Volkmann T. "ISS-EML batch #3 – science activities: PARSEC thermophysical properties." Investigators Working Group Meeting – IWG#23, ESA-ESTEC, Noordwijk Netherlands, May 22, 2024. Abstracts. Investigators Working Group Meeting – IWG#23, ESA-ESTEC, Noordwijk Netherlands, May 22, 2024. , May-2024
Articles in Peer-reviewed Journals	Nawer J, Stanford B, Kolbe M, Schneider S, Gossé S, Wunderlich RK, Mohr M, Borzì A, Neels A, Matson DM. "Thermodynamic assessment of evaporation during molten steel testing onboard the International Space Station." npj Microgravity. 2024 Jul 19;10(1):77. https://doi.org/10.1038/s41526-024-00416-1 , Jul-2024
Dissertations and Theses	Azersky C. "Evaluation of the influence of undercooling on defect structures using x-ray diffractometry." Dissertation, Tufts University, August 2024. , Aug-2024